

Assignment#1:

This assignment deals with computing the unsteady aerodynamic loads on a horizontal axis wind turbine. The DTU 10 MW reference wind turbine is used as example since the description of the machine including geometry and control is public.

Overall data for the DTU 10 MW WT:

Rotor radius $R = 89.17$ m

Number of blades 3

Rated power 10.64 MW

Cut in wind speed 4 m/s

Cut out wind speed 25 m/s

The blade is described with following table:
(for this turbine the table corresponds to 0 pitch)

r [m]	c [m]	β [deg]	t/c [%]
2,80	5,38	14,50	100,00
11,00	5,45	14,43	86,05
16,87	5,87	12,55	61,10
22,96	6,18	8,89	43,04
32,31	6,02	6,38	32,42
41,57	5,42	4,67	27,81
50,41	4,70	2,89	25,32
58,53	4,00	1,21	24,26
65,75	3,40	-0,13	24,10
71,97	2,91	-1,11	24,10
77,19	2,54	-1,86	24,10
78,71	2,43	-2,08	24,10
80,14	2,33	-2,28	24,10
82,71	2,13	-2,64	24,10
84,93	1,90	-2,95	24,10
86,83	1,63	-3,18	24,10
88,45	1,18	-3,36	24,10
89,17	0,60	-3,43	24,10

The airfoils used are named **FFA-W3-xxx**, where the last three digits indicate the profile thickness to chord in %. The following six airfoil tables are available, where cylinder_ds.txt corresponds to t/c=100%:

cylinder_ds.txt, FFA-W3-241_ds.txt, FFA-W3-301_ds.txt, FFA-W3-360_ds.txt,
FFA-W3-480_ds.txt and FFA-W3-600_ds.txt

The format in the airfoil data files is:

Each line specify the airfoil data as function of the angle of attack. The first column is the angle of attack, the next six columns contain the corresponding static lift coefficient, the

static drag coefficient, the static moment coefficient, the static dynamic stall coefficient, the linear lift coefficient and finally the fully stalled lift coefficient.

When estimating the lift and drag coefficients as function of the angle of attack at some radial position in your BEM code you need to interpolate in both angle of attack and thickness ratio t/c .

You can find the blade description in the file **bladedat.txt**. The first column gives radial positions, and the next three columns are the corresponding twist angle, chord and relative thickness.

Write an unsteady BEM code to investigate the pitch regulated DTU 10 MW reference wind turbine.

- First assume a constant rotational speed of $\omega=0.72$ rad/s, a constant wind speed of $V_o=8$ m/s, $\theta_{pitch}=0^\circ$ and the air density as 1.225 kg/m³ and plot as function of time thrust and power. Plot also the normal and tangential load distributions for the steady solution. Compare with results shown in the reference report for the DTU 10MW wind turbine, and perhaps your steady BEM code from 46300, if you have one.
- Next simulate a wind shear applying a wind shear exponent of $\nu=0.2$ and a wind speed of hub height $V_o(H)=8$ m/s. Still use $\omega=0.72$ rad/s and $\theta_{pitch}=0^\circ$. Plot on the same plot as function of time the thrust for each blade and the total thrust. Also plot as function of time the power. Explain the differences in the results from the previous question.
- Now remove the wind shear and consider again a constant wind speed of $V_o=8$ m/s and a rotational speed of $\omega=0.72$ rad/s, but use following prescribed pitch setting shown below. Explain the new time series for the power and the thrust and use your time history for the induced wind speed at $r=65.75$ m when explaining the results. Also compare your results with ASHES.

$$\begin{aligned}\theta_p(t) &= 0^\circ && \text{for } t < 100 \text{ s} \\ \theta_p(t) &= 2^\circ && \text{for } 100 \text{ s} \leq t \leq 150 \text{ s} \\ \theta_p(t) &= 0 && \text{for } t > 150 \text{ s}\end{aligned}$$

- Repeat the first question but with a turbulent wind. The pitch and rotational speed are kept constant at $\theta_{pitch}=0^\circ$ and $\omega=0.72$ rad/s. Make a turbulent box for the axial flow that covers the rotor for mean wind speed $V_o=8$ m/s corresponding to $z=119$ m. Plot for one blade the time history of the load normal to rotor plane, $p_z(r=65.75 \text{ m})$ and the total thrust and explain the power spectrum density of these signals, PSD(f).
- How do your results agree with ASHES